

Raja Senthilkumar

Frontend Developer · React.js · Next.js · MERN Stack

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PROFESSIONAL SUMMARY

Skilled and versatile **Full Stack Developer with 3+ years of hands-on experience** building and scaling modern web applications. Specializes in **React.js, Next.js, and the MERN stack** with proven capabilities in backend integrations, real-time analytics, and secure payment workflows. Comfortable leading end-to-end feature delivery — from database schema design and REST/GraphQL API development through to polished, component-driven frontends. Experienced with Agile/Scrum methodologies and modern authentication mechanisms including RBAC and Key cloak.

CORE TECHNICAL SKILLS

Frontend	React.js, Next.js, TypeScript, JavaScript (ES6+), Redux, Tailwind CSS, Styled Components, CSS3, Bootstrap
Backend	Node.js, Express.js, GraphQL (Apollo Server), REST APIs, WebSocket / Socket.IO
Databases	PostgreSQL, MySQL, MongoDB, Redis
Authentication	Key cloak (RBAC, SSO), JWT, AWS Cognito, Role-Based Access Control
Cloud & DevOps	AWS S3, Git, GitHub, Jira, Agile / Scrum
Third-Party APIs	Stripe (Billing & Subscriptions), Apache Superset, Konva.js (Canvas)

PROFESSIONAL EXPERIENCE

Software Developer

Oct 2024 – Apr 2026

Twilight IT Solutions Pvt Ltd

- Lead frontend development of a real-time Fleet Management System using React.js, Java, PostgreSQL, and Socket.IO — delivering live GPS tracking, geofencing alerts, and driver-behavior dashboards.
- Integrated Apache Superset with custom SQL queries for fuel analytics, time-series forecasting, and predictive cost analysis; embedded charts directly into the React app via REST API / iframe.
- Built and shipped key modules of Go Perla — a multi-tenant SaaS Document Analysis Platform — including Stripe subscription billing, AWS Cognito authentication, AWS S3 file storage, and a GraphQL API layer.
- Actively participated in Agile sprint ceremonies, conducted peer code reviews, and contributed to technical documentation for all shipped features.

Junior Software Developer

Apr 2023 – Sep 2024

App Xperts Private Limited

- Delivered the ULCMS e-Learning Management System end-to-end — React.js frontend, Node.js backend, PostgreSQL database, and Key cloak-based RBAC for three distinct user roles.
- Built a browser-based Video Editor using Konva.js with drag-and-drop canvas editing, layer management, object transformation, and a full undo-redo history stack.
- Implemented React Query for optimized data fetching, caching, and optimistic UI updates across both projects, eliminating redundant API calls and improving perceived performance.
- Created and maintained a shared reusable component library (forms, tables, modals, progress indicators) used across multiple internal projects.

Software Development Intern

Dec 2022 – Mar 2023

App Xperts Academy

- Assisted in frontend development using React.js, gained practical exposure to Node.js REST APIs, PostgreSQL queries, and Git-based collaborative workflows.
- Completed structured training in JavaScript (ES6+), React component architecture, and responsive UI design using CSS3 and Bootstrap.

EDUCATION

→ **Bachelor of Technology – Electronics & Communication Engineering** 2022 -passed out

KEY PROJECTS

Fleet Management System

Domain: Transportation / Telematics | **Stack:** React.js, Java, PostgreSQL, Konva.js, Socket.IO, Apache Superset

- Architected an end-to-end real-time fleet monitoring dashboard enabling live GPS path visualisation across 100+ vehicles simultaneously with sub-second WebSocket update latency.
- Implemented interactive geofence polygon drawing via Konva.js with server-side event triggers that dispatch instant push alerts whenever a vehicle enters or exits a defined zone.
- Designed a comprehensive telemetry ingestion pipeline capturing engine diagnostics (RPM, coolant temperature, DTC fault codes), driver behaviour scores, harsh-braking events, and idle-time metrics.
- Wrote advanced PostgreSQL queries using window functions, CTEs, and multi-table joins to compute cumulative fuel consumption, per-route cost breakdowns, and driver efficiency rankings over configurable date ranges.
- Integrated Apache Superset with custom SQL datasets aggregating fuel-level deltas, engine runtime, and idle periods; applied built-in time-series forecasting to project upcoming fuel costs and embedded all charts via REST API / iframe.
- Built a modular component library for event logs, engine-fault notifications, and historical route replay, each component scoped to its own data slice to minimise API payload size.

Go Perla – SaaS Document Analysis Platform

Domain: SaaS / Document Intelligence | **Stack:** React.js, Node.js, GraphQL, MongoDB, Redis, AWS Cognito, AWS S3, Stripe

- Designed a multi-tenant SaaS architecture with full per-organisation data isolation; built tenant-resolution middleware that routes every API request to the correct namespace using JWT claims issued by AWS Cognito.
- Engineered a secure document upload pipeline using pre-signed S3 URLs so files transfer directly from the browser to S3 without touching the application server, then trigger an async processing job on completion.
- Developed a GraphQL API layer with Apollo Server covering document ingestion, annotation, result querying, and team management; used DataLoader batching to eliminate N+1 fetch patterns against MongoDB.
- Integrated Stripe for multi-tier subscription plans (Starter, Pro, Enterprise) — including upgrade/downgrade proration, usage-metered billing per document page processed, webhook reconciliation, and automated PDF invoice delivery.
- Implemented a Redis caching layer for document metadata and permission lookups; optimised the React frontend with lazy loading, Suspense-based code splitting, and memoised selectors to improve Lighthouse performance scores.

ULCMS – E-Learning Management System

Domain: Education / E-Learning | **Stack:** React.js, Node.js, PostgreSQL, Keycloak, React Query

- Delivered a full-stack LMS with role-segregated dashboards for Super Admin, Instructor, and Student roles; access boundaries enforced via Keycloak realm roles and resource policies with JWT validation on every API route.
- Built a dynamic quiz engine supporting MCQ, true/false, and short-answer question types with configurable time limits, randomised question ordering, auto-grading, and instant score feedback on submission.
- Implemented course-progression tracking that records per-lesson completion status, calculates overall course percentage, and unlocks subsequent modules only after prerequisites are satisfied — enforced atomically via PostgreSQL triggers.
- Used React Query (TanStack Query) for all data-fetching with stale-time configuration, background refetch, and optimistic updates on quiz submissions to keep the UI responsive without manual loading spinners.
- Created a reusable component library (course cards, progress bars, quiz blocks, user-profile panels) with Tailwind CSS, maintaining visual consistency across 30+ screens and accelerating feature delivery across sprints.

Video Editor – Poster & Video Customisation Tool

Domain: Media & Design | **Stack:** React.js, Konva.js, PHP, PostgreSQL

- Built a fully browser-based visual editor using the Konva.js HTML5 Canvas API, enabling users to design custom posters and video overlays — supporting images, text blocks, shapes, and stacked background layers on a single canvas.
- Implemented a complete drag-and-drop interaction model where every canvas object is individually selectable, movable, resizable via corner handles, and rotatable with pixel-level snapping guides for precise alignment.
- Engineered a robust undo-redo history stack using the Command pattern — each user action (move, resize, add, delete, style change) is serialised as a reversible command object, supporting unlimited undo steps.
- Developed a layers panel allowing users to reorder, lock, hide, and group objects, mirroring the UX of professional design tools while keeping the React implementation lightweight and performant.
- Built the PHP backend with a PostgreSQL schema to persist template definitions, serialised canvas state (JSON), and user-uploaded asset references, with auto-save on every change and full project restoration on next login.